

CAdam: Context-Adaptive Moment Estimation for 3D Gaussian Densification in Generative Distillation

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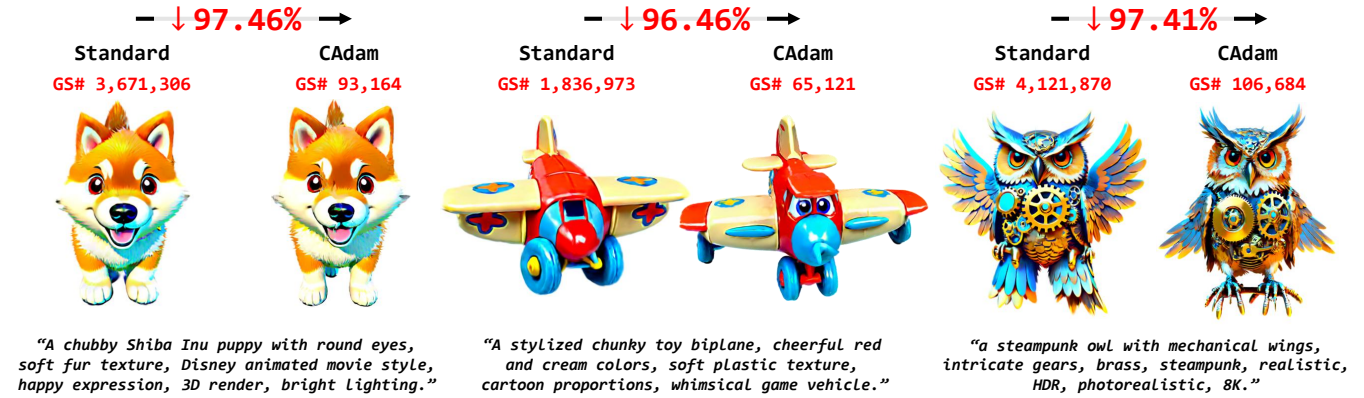


Fig. 1. CAdam reduces Gaussian primitives by up to 97% in optimization-based generative 3D Gaussian Splatting. Across diverse prompts and structures, our method achieves substantially more compact representations while maintaining overall comparable rendered appearance relative to standard densification.

Adaptive densification is the engine of 3D Gaussian Splatting (3DGS). However, when transposed to the optimization-based Generative Distillation paradigm, this reconstruction-native mechanism reveals fundamental limitations, resulting in inefficient representations cluttered with redundant primitives. We diagnose this failure as a *Densification Dilemma* stemming from the stochastic nature of generative guidance: the standard magnitude-based accumulation indiscriminately aggregates transient noise alongside geometric signals, making it difficult to strike a balance between over-densification and under-fitting. To resolve this, we introduce **Context-Adaptive Moment Estimation (CAdam)**, a novel framework that reinterprets densification as a statistically grounded signal verification problem. CAdam leverages the first moment of gradients to exploit the interference principle—where stochastic fluctuations cancel out via *destructive interference* while consistent geometric drifts accumulate via *constructive interference*—effectively disentangling the underlying signal from the generative noise floor. This is further augmented by a quantile-based context awareness and an intrinsic Signal-to-Noise Ratio (SNR) gating mechanism, which ensure robust

adaptation across optimization stages and enable the soft termination of densification. Extensive experiments across diverse objectives (SDS, ISM, VFDS) and strong generative 3DGS backbones show that CAdam reduces Gaussian count by 85%–97% relative to standard densification while preserving overall comparable perceptual quality. These results highlight signal-aware density control as a practical way to improve memory efficiency in optimization-based generative distillation.

CCS Concepts: • Computing methodologies → Rendering.

Additional Key Words and Phrases: 3D Gaussian Splatting, Generative Distillation, Gaussian Densification

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1 Introduction

Optimization-based generative 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023] has rapidly emerged as a powerful paradigm for text-to-3D synthesis. From early adaptations to recent high-performance systems, this optimization-based branch inherits the *Generative Distillation* paradigm. It “dreams up” 3D assets by distilling knowledge from pre-trained 2D diffusion [Rombach et al. 2022] or flow models [Lipman et al. 2022; Liu et al. 2022] into the 3D Gaussian space.

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Despite substantial progress in initialization strategies and generative loss designs, the mechanism governing structural evolution—densification—remains largely inherited from reconstruction-oriented pipelines, without explicit adaptation for generative supervision.

Densification critically determines both representational efficiency and geometric fidelity. However, existing generative 3DGS methods still rely on gradient-magnitude accumulation, a heuristic designed for reconstruction. This approach relies on a fundamental assumption: *gradient magnitude is a reliable proxy for geometric error*. In reconstruction tasks, the optimization targets fixed ground-truth views. As the rendered images converge to this reference, the gradient magnitudes naturally diminish. Thus, persistent high magnitudes reliably indicate under-reconstruction. In generative settings, however, supervision is provided by stochastic pseudo-targets induced by random timesteps, noise, and view sampling. As a result, gradients exhibit persistent non-convergence, even when the geometry is already plausible. Under these dynamics, magnitude accumulation fails to distinguish coherent geometric drift from stochastic fluctuations, leading to a **Densification Dilemma**: low thresholds cause unbounded proliferation of redundant Gaussians, while high thresholds suppress valid refinement and induce geometric under-fitting. This results in inefficient representations dominated by redundant primitives with little contribution to visual quality.

To resolve this dilemma, we introduce **Context-Adaptive Moment Estimation (CAdam)**, a densification framework that reinterprets densification not as a magnitude-driven heuristic, but as a statistical signal verification process. Inspired by moment estimation in adaptive optimizers [Kingma and Ba 2015], CAdam instills reliability into the densification process through three tightly coupled design principles. First, **Momentum-based Signal Verification** accumulates gradient vectors instead of scalar magnitudes, allowing stochastic noise to cancel via destructive interference while preserving coherent geometric drift. Second, **Context-Adaptive Selection** evaluates densification candidates using population-aware quantile ranking and an intrinsic Signal-to-Noise Ratio (SNR), ensuring that refinement is triggered only by statistically reliable signals. Finally, **Selective Structural Refinement** restricts topological updates to verified candidates, preventing redundant primitive proliferation and enabling efficient capacity allocation.

CAdam generalizes across optimization-based generative 3DGS settings, including Score Distillation Sampling (SDS), Interval Score Matching (ISM), and Vector Field Distillation Sampling (VFDS). Experiments demonstrate that CAdam reduces Gaussian counts by **85%–97%** compared to standard densification while maintaining overall comparable perceptual quality. We provide a visual comparison of this structural efficiency in Fig. 1.

Our contributions are summarized as follows:

- We diagnose a fundamental mismatch between reconstruction-oriented densification and the Generative Distillation paradigm, defining the **Densification Dilemma**: low thresholds trigger redundant primitive proliferation, whereas high thresholds suppress valid refinement.
- We propose **CAdam**, a statistically grounded density control framework. By verifying geometric signals via **Momentum** and employing a **Context-Adaptive Selection** strategy, we enable **Selective Structural Refinement** that strictly densifies only reliable regions.
- We demonstrate that CAdam serves as a unified solution across diverse generative 3DGS backbones. Extensive experiments confirm an **85%–97%** reduction in Gaussian primitives while maintaining overall comparable perceptual quality, substantially reducing structural redundancy.

2 Related Works

2.1 3D Generation and Distillation

Text-to-3D distillation optimizes 3D representations using generative priors of 2D diffusion models. Early NeRF-based methods established SDS [Poole et al. 2023; Wang et al. 2023a; Yu et al. 2023; Zhu et al. 2023] and improved its diversity [Wang et al. 2023b], and stability [Wu et al. 2024; Yan et al. 2025].

With the adoption of explicit primitives, 3DGS extended SDS to Gaussian representations [Chen et al. 2024b; Tang et al. 2023; Yi et al. 2024]. However, applying diffusion gradients independently across random noise levels exposes a limitation: lack of pathwise consistency, often leading to over-smoothed pseudo-targets and unstable multi-view optimization [Alldieck et al. 2024].

To mitigate these issues, recent work has shifted toward trajectory-based distillation. ODE-based methods leverage DDIM inversion [Liang et al. 2024] or latent consistency models [Zhong et al. 2024], while Flow Matching variants [Li et al. 2024a, 2025b; Lukoianov et al. 2024; Miao et al. 2024] construct deterministic trajectories that reduce the mean-seeking bias of standard SDS. Complementary efforts address multi-view inconsistency and prior-induced bias through consistency objectives or 3D-aware regularization [Jin et al. 2025; Li et al. 2024b; Miao et al. 2025; Zhang et al. 2025a,b; Zhou et al. 2025; Zhu et al. 2025]. In parallel, several extensions expand controllability and representation variants within generative 3DGS [Cai et al. 2024; Chen et al. 2024a; Di et al. 2025; He et al. 2024; Huang et al. 2024; Pham et al. 2024; Tran et al. 2024; Zhuo et al. 2024].

Orthogonal to recent feed-forward and native/structured-latent 3D methods targeting fast asset generation [Hu et al. 2025; Lai et al. 2025; Li et al. 2025a; Liu et al. 2024; Tang et al. 2024, 2025; Wu et al. 2025; Xiang et al. 2025, 2024; Xu et al. 2024; Zhang et al. 2024a], we focus on optimization-based generative distillation. We observe that stochastic gradient fluctuations in such settings fundamentally hinder effective Gaussian densification. Our work addresses this bottleneck with signal-aware densification that aligns primitive growth with the statistical properties of generative supervision.

2.2 Densification and Structural Dynamics

Adaptive density control governs Gaussian cloning and splitting during 3DGS optimization. While effective in reconstruction, its heuristic nature often results in over- or under-densification, motivating refinements to densification criteria and operations.

Several works improve gradient-based densification by enforcing directional consistency or locally adaptive thresholds, leading to more stable density evolution [Fan et al. 2024; Ye et al. 2024a; Zhang et al. 2024b]. Others improve densification operations with more controllable and temporally stable update rules based on residual formulations or momentum-inspired mechanisms [Kotovenko

et al. 2025; Lyu et al. 2025; Yuan et al. 2025]. Other approaches incorporate additional cues, such as color and frequency information [Kim et al. 2024; Li et al. 2025c], or explore stochastic and sparsity-oriented formulations, including probabilistic sampling and fragment-level pruning [Kheradmand et al. 2024; Ye et al. 2024b]. Recent post-pruning methods further reduce redundant primitives after or alongside representation growth, using uncertainty scores or learnable pruning masks [Hanson et al. 2025; Zhang et al. 2024c].

All of these methods are developed under deterministic reconstruction assumptions. In contrast, we address densification in the stochastic setting of optimization-based generative distillation.

3 Preliminaries

3.1 Densification in 3D Gaussian Splatting

3DGS framework [Kerbl et al. 2023] represents a scene as a set of anisotropic 3D Gaussians G_i , each parameterized by a mean position μ_i , covariance Σ_i , opacity α_i , and spherical harmonics (SH) coefficients. Since 3DGS begins with a deliberately coarse initialization for efficiency and stability, it relies on adaptive *densification* to progressively increase representational capacity.

Densification candidates are identified through the accumulation of view-space positional gradients over a fixed window of K optimization steps. Let $\mathbf{g}_{i,t} = \nabla_{\mu_i^{2D}} \mathcal{L}$ denote the view-space positional gradient of Gaussian G_i at step t . The accumulated gradient statistic is defined as

$$\mathcal{A}_i = \frac{1}{K} \sum_{t=1}^K \|\mathbf{g}_{i,t}\|_2, \quad (1)$$

which serves as a proxy for persistent positional instability. If $\mathcal{A}_i$ exceeds a predefined threshold τ_{pos} , the Gaussian is selected for densification. The specific densification operation is determined by its scale S_i :

$$\text{Action}(G_i) = \begin{cases} \text{Split,} & \text{if } \mathcal{A}_i > \tau_{pos} \text{ and } \|S_i\|_\infty > \tau_{scale} \\ \text{Clone,} & \text{if } \mathcal{A}_i > \tau_{pos} \text{ and } \|S_i\|_\infty \leq \tau_{scale} \end{cases}. \quad (2)$$

To maintain numerical stability and prevent degenerate primitives, Gaussians with negligible opacity or excessively large scales are subsequently pruned.

This strategy relies on a fundamental assumption of reconstruction: the positional gradient norm is a reliable indicator of the *photometric error*:

$$\mathcal{L}_{photo} = (1 - \lambda) \|\hat{I} - I_{GT}\|_1 + \lambda \text{D-SSIM}(\hat{I}, I_{GT}). \quad (3)$$

where $\hat{I}$ is a rendered image and I_{GT} is a ground-truth image. As reconstruction progresses, gradients for well-represented regions naturally decay, while persistently large gradients reliably indicate under-reconstruction.

3.2 Gradient Formulation in Generative Distillation

In contrast to reconstruction, text-to-3D generation optimizes a 3D representation θ by distilling knowledge from a pretrained 2D generative prior. Most distillation-based objectives, ranging from Score Distillation Sampling (SDS) to recent advances like Interval Score Matching (ISM) and Vector Field Distillation Sampling (VFDS), can be unified into a general *pseudo-target alignment* framework.

Specifically, the update direction aligns the latent representation $x_0 := g(\theta, c)$ of $\hat{I}$ with the denoised target $\hat{x}_0^t$ estimated from the noisy latent x_t . Here, c denotes the rendering conditions, including the camera pose and intrinsic parameters under which the image is rendered. Incorporating a timestep-dependent weighting function $w(t)$ and the domain scaling factor $\gamma(t) = \frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}}$, the gradient is formulated as:

$$\nabla_\theta \mathcal{L}_{\text{SDS}} \approx \mathbb{E}_{t,\epsilon,c} \left[\frac{w(t)}{\gamma(t)} (x_0 - \hat{x}_0^t) \frac{\partial g(\theta, c)}{\partial \theta} \right]. \quad (4)$$

Crucially, the discrepancy term $(x_0 - \hat{x}_0^t)$ admits equivalent parameterizations under linear diffusion dynamics [Li and He 2025]:

$$(x_0 - \hat{x}_0^t) \propto (\hat{\epsilon}_\phi - \epsilon) \propto (\hat{v}_\phi - v_t). \quad (5)$$

where $\hat{\epsilon}_\phi$ and $\hat{v}_\phi$ represent the noise and velocity predicted by the pre-trained model, respectively. This equivalence demonstrates that various guidance strategies share a common objective: minimizing the error between the current state and a predicted target on the learned data manifold.

However, unlike reconstruction, this supervision is inherently *stochastic*. Each update relies on single-step denoising at randomly sampled timesteps and noise realizations, producing gradients with high variance and no notion of convergence to a fixed target. This stochasticity persists even in interval-based variants and is further amplified by view sampling and manifold misalignment during early optimization. As a result, gradient magnitudes fluctuate persistently over time, violating the assumptions under which reconstruction-driven densification remains reliable.

4 Method

We introduce **CAdam**, a densification framework that reinterprets 3D Gaussian densification as a problem of statistical signal verification, rather than heuristic accumulation of gradient magnitudes.

Before detailing CAdam, we formalize why the standard densification mechanism—originally designed for reconstruction—leads to a fundamental failure mode in generative distillation.

4.1 Problem Definition

Standard densification strategy relies on the accumulation of positional gradient magnitudes, implicitly assuming that gradients decay as optimization progresses. While valid for reconstruction, we demonstrate that it is fundamentally violated in generative distillation due to its non-convergent and stochastic supervision.

Convergence in Reconstruction. In reconstruction, optimization is driven by a static objective defined over fixed ground-truth views. The rendered image $\hat{I}$ converges to the target image I_{GT} , and the photometric loss $\mathcal{L}_{photo}$ naturally diminishes. Consequently, the gradients decay over time ($\lim_{t \rightarrow \infty} \|\nabla_\mu \mathcal{L}\| = 0$). Although random view sampling introduces mild stochasticity, the gradients still persistently point toward a single optimum. In this regime, accumulating large gradient norms reliably identifies regions of geometric under-fitting.

Instability in Generative Distillation. Generative distillation replaces the static ground truth with a stochastic pseudo-target $\hat{x}_0^t$, obtained via single-step denoising at randomly sampled timesteps t and

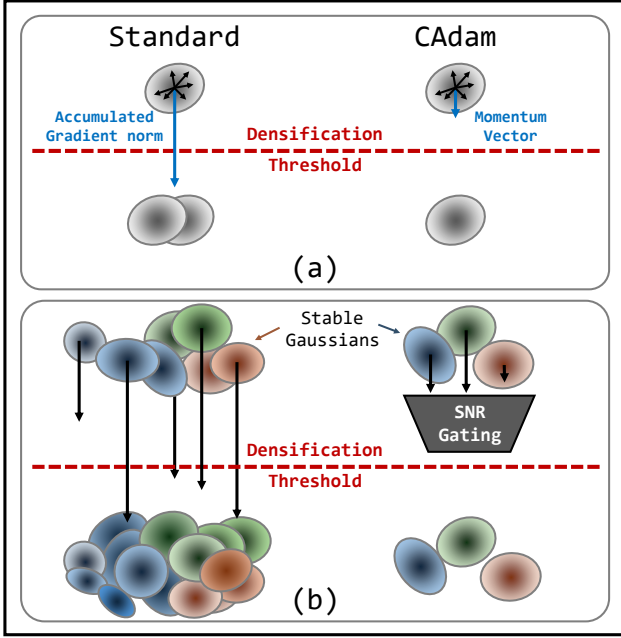


Fig. 2. Conceptual comparison between magnitude-based densification and CAdam. (a) Standard densification uses the accumulated gradient norm as a scalar trigger, whereas CAdam relies on the momentum vector. (b) For geometrically stable Gaussians, standard accumulation repeatedly exceeds the densification threshold, leading to unnecessary splitting. CAdam suppresses this behavior through SNR gating.

noise realizations ϵ . The optimization target therefore changes at every iteration, including two distinct dynamics: (1) **Non-convergence**: The alignment error ($x_0 - \hat{x}_0^t$) remains inherently non-zero, resulting in persistent gradient magnitudes even when the geometry is visually plausible. (2) **Directional Variance**: Variance in the noise prediction $\hat{\epsilon}_\phi$ causes rapid fluctuations in the update direction. We refer to this high-frequency oscillatory behavior—where stochastic guidance yields conflicting gradient signals across steps—as the *zigzag phenomenon*.

Densification Dilemma. The failure arises when reconstruction-oriented densification is applied under these stochastic dynamics. Let $\mathbf{g}_t$ denote the gradient of each Gaussian G_i at step t , modeled as the superposition of a deterministic geometric drift $\bar{\mathbf{g}}_t$ and a stochastic noise component ξ_t :

$$\mathbf{g}_t = \bar{\mathbf{g}}_t + \xi_t, \quad \text{where } \mathbb{E}[\xi_t] = \mathbf{0}, \quad (6)$$

where $\bar{\mathbf{g}}_t$ represents the coherent update direction for structural refinement and ξ_t captures the stochastic fluctuations driving the *zigzag phenomenon*.

Ideally, densification should respond only to $\|\bar{\mathbf{g}}_t\|$. However, due to the convexity of the L_2 norm, the standard metric inherently entangles the signal with the noise magnitude:

$$\mathcal{A}_{naive} = \frac{1}{K} \sum_{t=1}^K \|\bar{\mathbf{g}}_t + \xi_t\| \approx \|\bar{\mathbf{g}}_t\| + \frac{1}{K} \sum_{t=1}^K \|\xi_t\|. \quad (7)$$

Even when the geometry is plausible ($\bar{\mathbf{g}}_t \approx \mathbf{0}$), the accumulated noise floor $\frac{1}{K} \sum \|\xi_t\|$ remains significantly non-zero.

This leads to a **densification dilemma**: a low threshold captures fine details but is dominated by noise, triggering unbounded proliferation of redundant Gaussians; a high threshold suppresses noise but prevents valid refinement, yielding coarse or incomplete geometry. Crucially, no fixed threshold can resolve this ambiguity.

To resolve this limitation, we must shift densification paradigm from “magnitude-based detection” to “statistical signal verification”, via moment estimation to cancel out ξ_t and recover $\bar{\mathbf{g}}_t$.

4.2 Momentum-based Signal Verification

To resolve the densification dilemma, we interpret momentum not as an optimization aid, but as a mechanism for verifying coherent geometric signals (see Fig. 2). We leverage the first-moment estimate from Adam [Kingma and Ba 2015] to disentangle the coherent geometric drift $\bar{\mathbf{g}}_t$ from the stochastic noise ξ_t .

Unlike magnitude-based accumulation, we track positional gradients directly in world space ($\mathbf{g}_t = \nabla_{\mu_{xyz}} \mathcal{L}$), removing projection-dependent fluctuations from random camera sampling. The momentum vector $\mathbf{m}_t$ is updated via an exponential moving average (EMA) and bias-corrected to obtain an unbiased estimate $\hat{\mathbf{m}}_t$:

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_t, \quad \hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1 - \beta_1^t}, \quad (8)$$

where β_1 is the decay rate. For newly instantiated Gaussians, we use each Gaussian’s age as the step count t for bias correction, rather than the global iteration count.

This vector-based integration exploits a key property of generative guidance: stochastic noise is zero-mean ($\mathbb{E}[\xi_t] = \mathbf{0}$). Then, high-frequency fluctuations (ξ_t) undergo *destructive interference* and cancel out towards zero. Conversely, the consistent geometric drift $\bar{\mathbf{g}}_t$ accumulates via *constructive interference*, resulting in a stable direction vector.

Consequently, the unbiased momentum effectively extracts the underlying geometric signal ($\hat{\mathbf{m}}_t \approx \bar{\mathbf{g}}_t$), yielding a robust densification criterion. Even in fine-detail regions where gradient magnitudes are small, directional consistency is preserved, allowing valid refinement to be detected where magnitude-based heuristics fail.

4.3 Context-Adaptive Selection

Identifying valid densification candidates requires more than the absolute magnitude of momentum $\hat{\mathbf{m}}_i$. As gradient scales decay over optimization and generative guidance remains stochastic, we propose a **Context-Adaptive Selection** strategy that jointly considers *relative importance* within the scene and *absolute reliability* of each signal.

Quantile-based Selection. In response to shifting gradient scales, we adaptively select Gaussians for densification based on their relative importance within the scene. A dynamic threshold derived from population statistics is employed to identify candidate G_i . Specifically, G_i is considered for densification if it satisfies:

$$\|\hat{\mathbf{m}}_i\| > \text{Quantile} \left(\{\|\hat{\mathbf{m}}_j\|\}_{j=1}^N, \tau_Q \right), \quad (9)$$

where τ_Q is the target percentile (e.g., 0.9).

This strategy remains effective throughout optimization. In early stages, it prioritizes coarse structural corrections with large momentum, while in later stages—when global gradients diminish—it continues to surface fine-detail regions whose momentum, though small in absolute value, remains significant compared to stabilized areas. This quantile-based approach naturally induces an automated *coarse-to-fine* refinement schedule without requiring a hand-tuned threshold.

SNR-based Selection. Momentum identifies directional trends but does not quantify signal confidence. To verify whether a candidate represents a reliable geometric signal rather than accumulated noise, we introduce a **reliability condition** based on the Signal-to-Noise Ratio (SNR). We leverage the second raw moment $\mathbf{v}_t$ from Adam [Kingma and Ba 2015]:

$$\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2)(\mathbf{g}_t \odot \mathbf{g}_t), \quad \hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1 - \beta_2^t}. \quad (10)$$

We define the Intrinsic SNR as

$$\text{SNR}_t = \frac{\|\hat{\mathbf{m}}_t\|_2}{\sqrt{\|\hat{\mathbf{v}}_t\|_1 + \epsilon}}. \quad (11)$$

measuring the strength of coherent drift relative to stochastic fluctuations. A Gaussian is selected for densification only if $\text{SNR}_t > \tau_{\text{SNR}}$.

This metric effectively distinguishes signal from noise; Gaussians G_i dominated by stochastic fluctuations exhibit small $\|\hat{\mathbf{m}}_i\|$ due to *destructive interference* but large $\|\hat{\mathbf{v}}_i\|$ due to magnitude accumulation, resulting in a low SNR.

Densification is finalized by the intersection of quantile-based ranking and SNR criteria. As geometry converges ($\bar{\mathbf{g}}_t \rightarrow \mathbf{0}$), the coherent signal in the momentum vanishes, whereas stochastic fluctuations continue to accumulate in the variance term $\hat{\mathbf{v}}_t$. This naturally drives the SNR below τ_{SNR} even for top-quantile Gaussians. This intersection acts as a **soft termination mechanism**, ceasing densification once no statistically significant geometric drift remains and preventing redundant primitives.

4.4 Selective Structural Refinement

We decouple *when* to densify from *how* to modify geometry. The densification trigger is governed solely by statistical consistency (Momentum and SNR), while the structural operation (Clone vs. Split) follows the standard spatial scale criterion.

Action Criteria. Once a Gaussian is selected for densification ($M_{\text{densify}} = \text{True}$), the structural adjustment is determined as follows:

$$\text{Action} = \begin{cases} \text{Split,} & \text{if } \|S_i\|_\infty > \tau_{\text{scale}} \\ \text{Clone,} & \text{otherwise} \end{cases} \quad (12)$$

This separation ensures that temporal signal verification is orthogonal to spatial resolution control, allowing CAdam to be seamlessly integrated with existing 3DGS structural heuristics without introducing new geometric hyperparameters.

Selective Opacity Reset. The standard practice of periodically resetting global opacity imposes a dilemma: it indiscriminately degrades plausible geometries and introduces artificial gradient spikes that corrupt momentum history (EMA) with non-geometric transients. To mitigate this, we introduce a **Selective Opacity Reset**, targeting

only unreliable primitives identified by low Intrinsic SNR after an initial warm-up:

$$\alpha_{\text{new}} = \begin{cases} 0.01, & \text{if } \text{SNR} < \tau_{\text{SNR}} \\ \alpha_{\text{old}}, & \text{otherwise} \end{cases} \quad (13)$$

Noise-dominated primitives naturally exhibit low SNR and are selectively suppressed, facilitating their removal during pruning. This targeted reset ensures a balance between stability (preserving established structures) and plasticity (removing noisy artifacts to free up capacity for valid geometry), maintaining the statistical integrity of momentum statistics throughout training.

5 Experiments

5.1 Experimental Setup

Baseline and Implementation. We compare CAdam against the standard 3DGS densification [Kerbl et al. 2023], which is the default density control used by most generative pipelines. To assess generality, CAdam is integrated into multiple representative backbones and distillation objectives. For controlled analyses of optimization behavior and ablations, we use a representative optimization-based 3DGS pipeline with VFDS loss and Stable Diffusion 3.5. All experiments are run on a single NVIDIA RTX A5000 (24GB). Unless specified, we use a fixed set of hyperparameters for CAdam to demonstrate robustness: quantile threshold $\tau_Q = 0.9$, SNR threshold $\tau_{\text{SNR}} = 0.1$, and momentum decay $\beta_1 = 0.9$. CAdam keeps each backbone’s original K -step densification schedule: moment/SNR states are updated online at every optimization step, while split/clone candidates are selected only at scheduled densification calls using the quantile-SNR gate. The selective opacity reset schedule is also retained. See the supplementary material for implementation details and pseudocode.

Evaluation Protocol. We evaluate CAdam on diverse text prompts from established benchmarks such as T3Bench [He et al. 2023] and the DreamFusion Gallery [Poole et al. 2023], supplemented by custom prompts generated via Gemini-Pro and GPT-4. From an initial pool of 400 prompts spanning three categories of *Material* (e.g., fur, glass), *Structure* (e.g., complex architecture), and *Concept* (e.g., fantasy scenes), we sample 50 prompts with a balanced distribution.

We organize evaluation along two complementary axes. First, we assess *visual fidelity* using three criteria: (1) **Semantic Alignment**, measured by CLIP Score [Hessel et al. 2021] and ImageReward [Xu et al. 2023], for prompt consistency; (2) **Perceptual Aesthetics**, evaluated via HPS v2 [Wu et al. 2023], reflecting human visual preference; and (3) **Cleanliness and Naturalness**, assessed by NIQE [Mittal et al. 2012], where lower values indicate fewer visual artifacts and noise. Second, we evaluate *efficiency*, measured by the *Number of Gaussians* (N_g) and *Storage Size* (MB). To complement the evaluation with geometric evidence, depth and multi-view visualizations are provided in the supplementary material.

5.2 Comparative Evaluation

We compare CAdam against standard densification strategy used in optimization-based generative 3DGS pipelines. As shown in Fig. 3, CAdam produces visually detailed 3D assets while using substantially fewer primitives than the baseline.

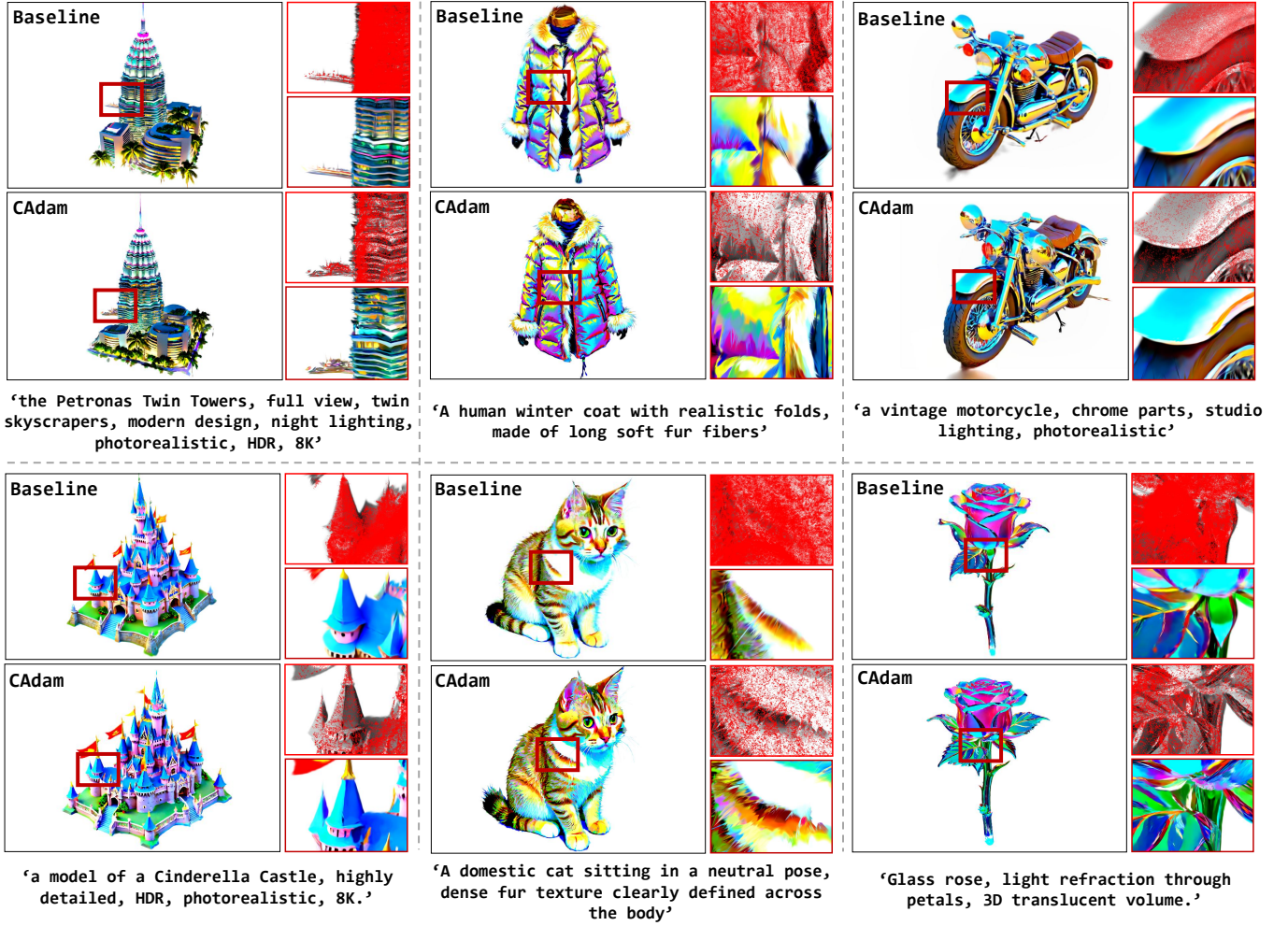


Fig. 3. Qualitative comparison between standard densification (Baseline) and CAdam across diverse prompts and structures. Zoomed-in regions highlight that baseline densification produces smoothed, blurry overlap patterns, whereas CAdam yields cleaner, more compact representations.

Generalization and Visual Fidelity. We validate the generalizability of CAdam across different distillation objectives and generative backbones. Fig. 4 shows that CAdam consistently improves efficiency under SDS, ISM, and VFDS. Fig. 5 further demonstrates consistent behavior across GaussianDreamer [Yi et al. 2024], LucidDreamer [Liang et al. 2024], GCS-BEG [Li et al. 2024b], and FlowDreamer [Li et al. 2024a]. Across all methods, CAdam significantly reduces primitive counts while maintaining comparable visual fidelity, demonstrating that its gains are model-agnostic.

Quantitative Compactness and Fidelity. Tab. 1 summarizes performance across GaussianDreamer (SDS), LucidDreamer (ISM), and FlowDreamer (VFDS). Across all backbones, CAdam significantly reduces the number of Gaussians by 85%–97%, yielding proportional storage savings compared to standard densification. Despite this substantial compression, quality metrics remain broadly comparable: CLIP and NIQE fluctuate across settings, while ImageReward improves under all objectives and HPS v2 remains comparable. These

Table 1. **Quantitative Study Results.** We evaluate CAdam on representative frameworks corresponding to each objective: GaussianDreamer (SDS), LucidDreamer (ISM), and FlowDreamer (VFDS).

Objective	Method	Quality Metrics ↑				Efficiency ↓	
		CLIP	ImgRwd	HPS v2	NIQE (↓)	# Gaussians	Storage
SDS	Baseline	30.14	-0.125	0.194	9.908	1067K	46.4MB
	CAdam	29.95	-0.028	0.185	9.643	143K	6.1MB
ISM	Baseline	31.10	0.413	0.225	6.438	1706K	110.6MB
	CAdam	28.48	0.518	0.225	8.167	81K	5.3MB
VFDS	Baseline	29.88	0.436	0.235	7.183	3373K	218.8MB
	CAdam	29.21	0.486	0.237	8.891	82K	5.3MB

results support CAdam as a compactness-oriented plug-and-play densification module that substantially reduces redundancy while maintaining broadly comparable quality.

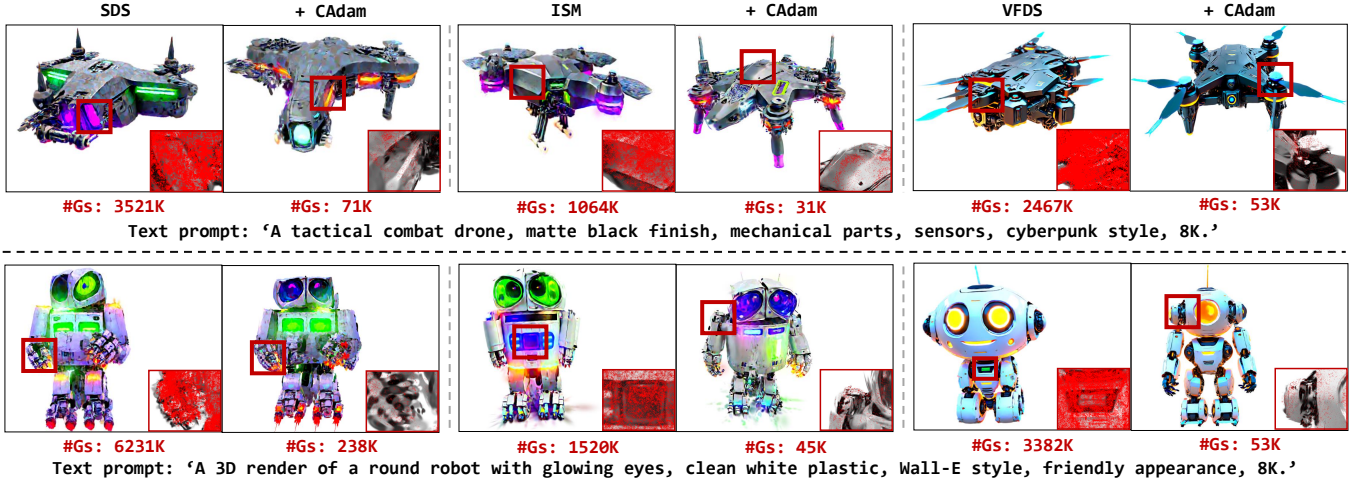


Fig. 4. Generalization across distillation objectives. CAdam consistently reduces Gaussian primitives across SDS, ISM, and VFDS. Insets highlight that efficiency gains stem from removing representational redundancy rather than degrading geometric detail.

Table 2. **User Study Results.** User preference rates (%) in a 2AFC setting. Results are averaged across samples from GaussianDreamer, LucidDreamer, and FlowDreamer. Metrics include Text Alignment (TA), Texture Detail (TD), Geometric Plausibility (GP), Cleanliness (CL), and Overall Preference (OP).

Method	TA	TD	GP	CL	OP
Baseline	50.5%	45.8%	51.9%	55.7%	50.4%
CAdam (Ours)	49.4%	54.1%	48.0%	44.3%	49.5%

Human Perception and Preference. We further assess perceptual quality via a blinded 2AFC study with 30 participants, comparing rendered videos across all frameworks. As shown in Tab. 2, CAdam achieves preference rates comparable to the dense baseline. The near-balanced overall preference rate of 49.5% for CAdam versus 50.4% for the baseline supports our main claim: CAdam achieves substantial compactness gains while maintaining comparable perceived quality.

5.3 Analysis of Training Dynamics

Fig. 6 validates the effectiveness of CAdam’s signal verification. Gradient analysis (top-left) shows that standard accumulation is dominated by noisy fluctuations and decays toward the noise floor, whereas CAdam preserves a more persistent structural signal through momentum-based verification. The primitive-count curve (bottom-left) shows that the baseline exhibits explosive growth and eventually reaches out-of-memory (OOM), while CAdam saturates at a much smaller count. The target masking maps (right) show a transition from coarse shape-level activation to more selective detail-oriented activation, supporting CAdam’s soft termination behavior.

5.4 Ablation Study

We analyze the contribution of each component in CAdam and its sensitivity to the quantile and SNR thresholds (Figs. 7 and 8).

Impact of Individual Components. Fig. 7 shows two representative failure modes that reveal the cumulative role of CAdam’s components. A momentum-only criterion suppresses part of the stochastic variation but fails to regulate late-stage primitive growth, eventually leading to OOM. Adding **Context-Adaptive Selection** reduces this growth, but without **Selective Opacity Reset** the representation becomes progressively over-suppressed and eventually vanishes. These results show that **Momentum-based Signal Verification** stabilizes noisy updates, **Context-Adaptive Selection** filters candidates by relative importance and reliability, and **Selective Opacity Reset** yields the most compact stable configuration by suppressing only low-SNR primitives.

Sensitivity and Controllability. Fig. 8 shows three representative threshold regimes: under-densification, balanced densification, and over-densification. Higher thresholds act as an aggressive filter and can under-densify weakly supported fine structures, while lower thresholds admit excessive primitive growth and denser representations. The default setting provides a balanced operating point in our experiments, indicating that the interplay of τ_Q and τ_{SNR} offers a practical control knob over the compactness-quality trade-off.

6 Conclusion

Optimization-based generative 3DGS has advanced rapidly, but its densification remains tied to deterministic reconstruction. Under stochastic generative supervision, magnitude accumulation conflates noise with geometric deficiency—the *Densification Dilemma*—causing over-growth or under-refinement. **CAdam** reframes densification as statistical decision-making: momentum verifies coherent structural signals, while context-adaptive quantile and intrinsic-SNR gates select reliable candidates, yielding self-regulating densification that terminates as structural signals vanish.

Across diverse objectives and backbones, CAdam reduces Gaussian primitives by 85%–97% while maintaining overall comparable perceptual quality. These results suggest that a substantial portion of

primitive growth in optimization-based generative 3DGS can be reduced by aligning density evolution with the statistics of generative supervision. Note that CAdam is not threshold-free: as illustrated in Fig. 8, overly loose thresholds may approach baseline-style over-densification, while overly aggressive thresholds can under-densify weakly supported fine structures such as thin parts or low-contrast details. In our experiments, the default setting provides a stable operating regime across diverse scenarios; remaining failures mainly arise when the underlying generative backbone provides inconsistent semantic or multi-view signals, which is orthogonal to our focus on density control during optimization.

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Fig. 5. Model-agnostic generalization of CAdam across optimization-based generative 3DGS frameworks. Qualitative comparisons on four generative backbones show that CAdam maintains comparable visual fidelity while dramatically reducing Gaussian counts. CAdam’s efficiency gains are robust and independent of the underlying backbone.

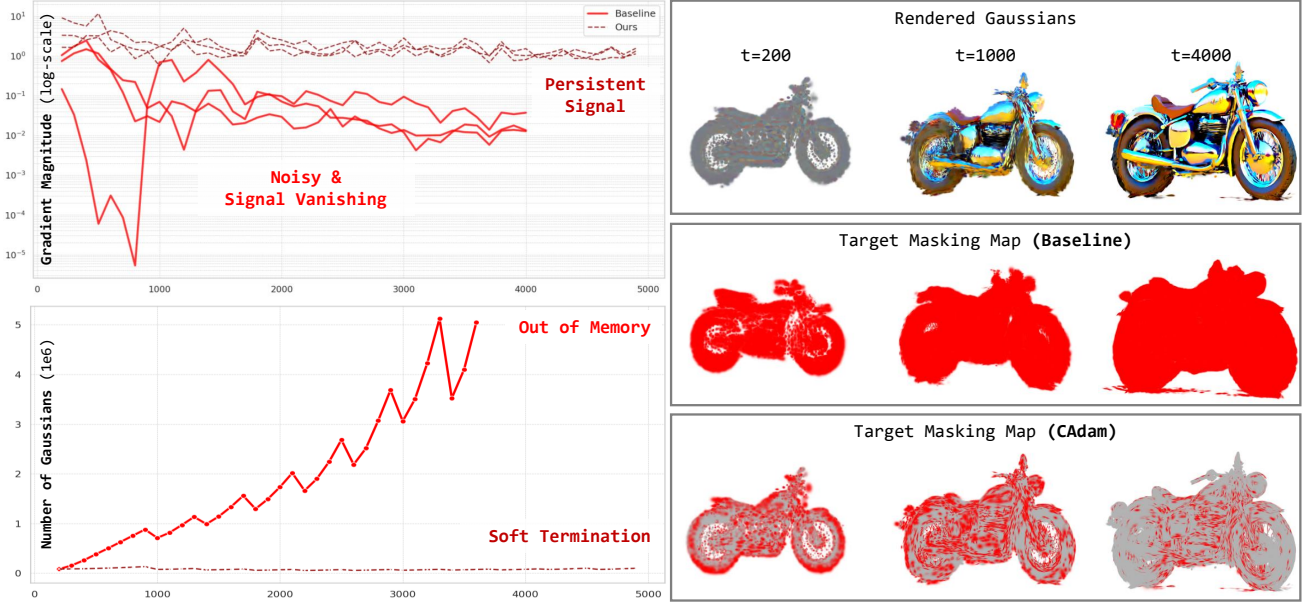


Fig. 6. Training dynamics under stochastic generative supervision. Top-left: Gradient magnitude analysis (log scale), tracked for a representative surviving Gaussian, shows that CAdam’s momentum-based signal verification maintains persistent structural magnitude to preserve history. Bottom-left: This signal disentanglement directly governs density evolution. Right: Visualization of densification candidates shows that CAdam selectively concentrates refinement on regions of genuine geometric deficiency.

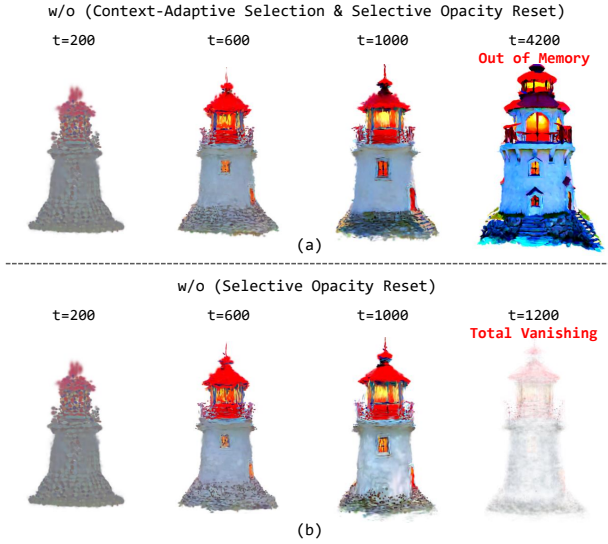


Fig. 7. Ablation study using the prompt “lighthouse, full view, smooth white cylindrical tower, detailed red lantern room, HDR, photorealistic, 8K.” (a) A momentum-only criterion cannot regulate primitive growth and eventually reaches OOM. (b) Removing Selective Opacity Reset causes progressive over-suppression and eventual vanishing.

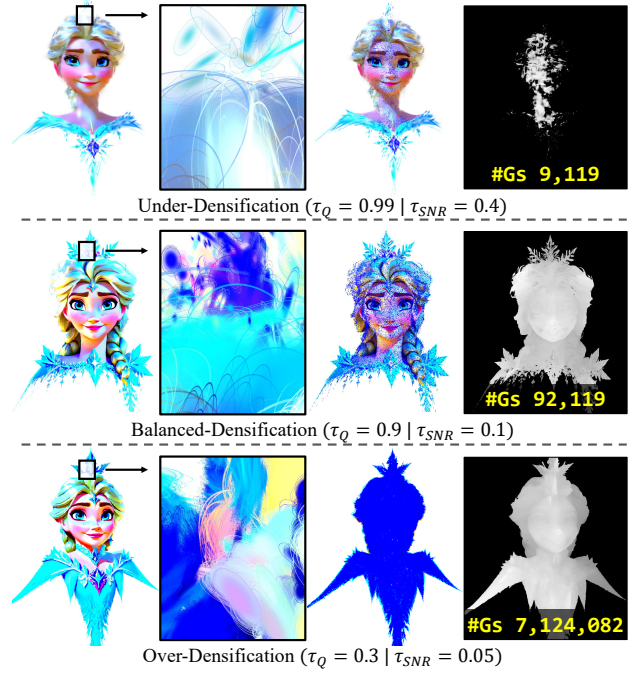


Fig. 8. Sensitivity analysis of the quantile and SNR thresholds. The rows show three representative regimes: under-densification, balanced densification, and over-densification. From left to right, we show the rendering, a zoomed-in edge visualization of the boxed region, the point-cloud density map, and the depth map with the final Gaussian count.

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